

N HECATE STRAIT BUOY 46183, CanadaWMO: **996460**Lat: **53.620N**Lon: **131.100W**Elev: **3**StdP: **101.29**Time Zone: **-9.00 (W09)**Period: **01-18**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.2	-0.1	N/A	N/A	N/A	N/A	N/A	N/A	19.7	6.6	18.4	6.8	7.3	20	0.839

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.6	16.2	N/A	15.6	N/A	15.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.2	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	16.5	14.8	-4.3	17.4	3.7	1.6	-6.9	18.6	-9.1	19.5	-11.2	20.3	-13.8	21.5		
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	6.0	5.6	5.6	7.0	9.1	11.4	13.2	13.9	12.5	10.3	7.5	6.0
	DBStd	3.50	2.59	2.09	2.15	1.31	1.31	1.36	1.19	1.04	0.98	1.38	2.12	2.35
	HDD10.0	726	124	124	138	92	33	2	0	0	0	13	77	123
	HDD18.3	3396	382	358	396	341	285	208	159	136	173	250	326	381
	CDD10.0	371	0	0	0	0	6	44	99	122	77	21	1	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.3	8.8	8.1	8.4	7.4	6.1	6.1	5.7	5.3	6.4	7.6	8.6	9.0
Precipitation	PrecAvg	1778	209	153	147	113	78	64	59	80	120	237	233	243
	PrecMax	2120	340	292	216	178	163	114	125	186	248	421	433	501
	PrecMin	1480	119	57	88	64	29	29	24	24	29	113	135	112
	PrecStd	186	56	60	39	28	35	23	26	40	47	73	64	81
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	10.0	10.0	11.3	13.0	16.4	16.6	17.3	15.7	13.5	11.7	10.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	9.7	9.1	9.2	9.9	11.9	14.8	15.7	16.5	14.9	12.7	10.7	9.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	9.2	8.4	8.6	9.2	11.2	14.0	15.2	15.9	14.4	12.3	10.4	9.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	8.6	8.0	8.0	8.7	10.7	13.3	14.9	15.4	14.0	11.9	10.0	8.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	2.2	2.1	2.2	1.8	1.6	1.6	1.6	1.6	1.5	1.7	2.3	2.2
		MCDBR	2.2	2.2	2.2	2.1	2.1	2.1	2.0	2.2	1.9	1.9	2.5	2.3
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.312	0.320	0.345	0.380	0.377	0.373	0.371	0.370	0.367	0.346	0.323	0.305	
	taud	2.405	2.491	2.434	2.326	2.399	2.450	2.496	2.515	2.484	2.445	2.339	2.326	
	Ebn at Noon	664	792	840	849	872	877	873	854	808	740	627	581	
	Edh at Noon	55	69	89	112	110	106	99	92	85	71	58	49	
All-Sky Solar Radiation	RadAvg	0.55	1.39	2.51	3.95	4.99	5.21	4.88	4.17	2.81	1.57	0.73	0.45	
	RadStd	0.07	0.22	0.28	0.22	0.43	0.37	0.51	0.33	0.29	0.20	0.09	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A